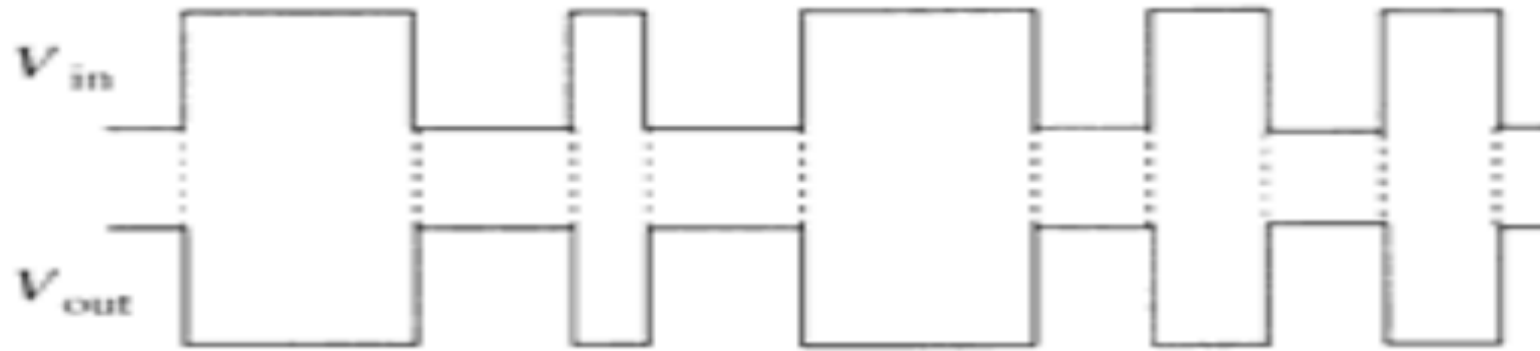


The Answer of Sheet 3

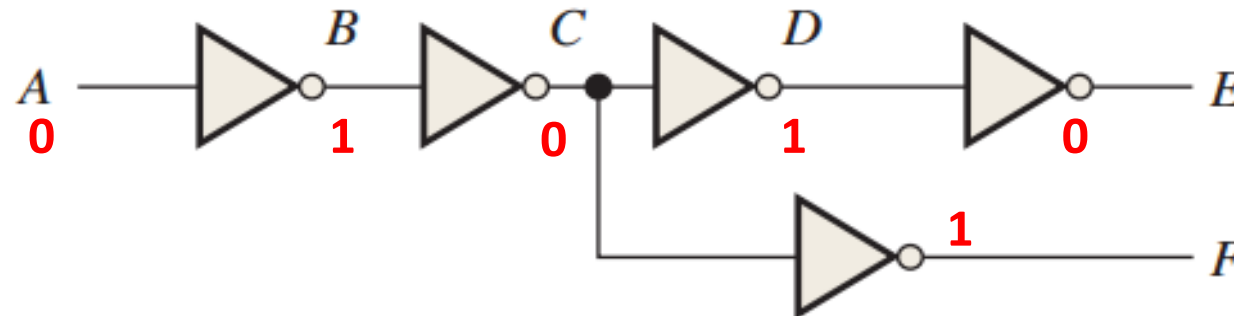
Logic Gates

Section 3-1 The Inverter

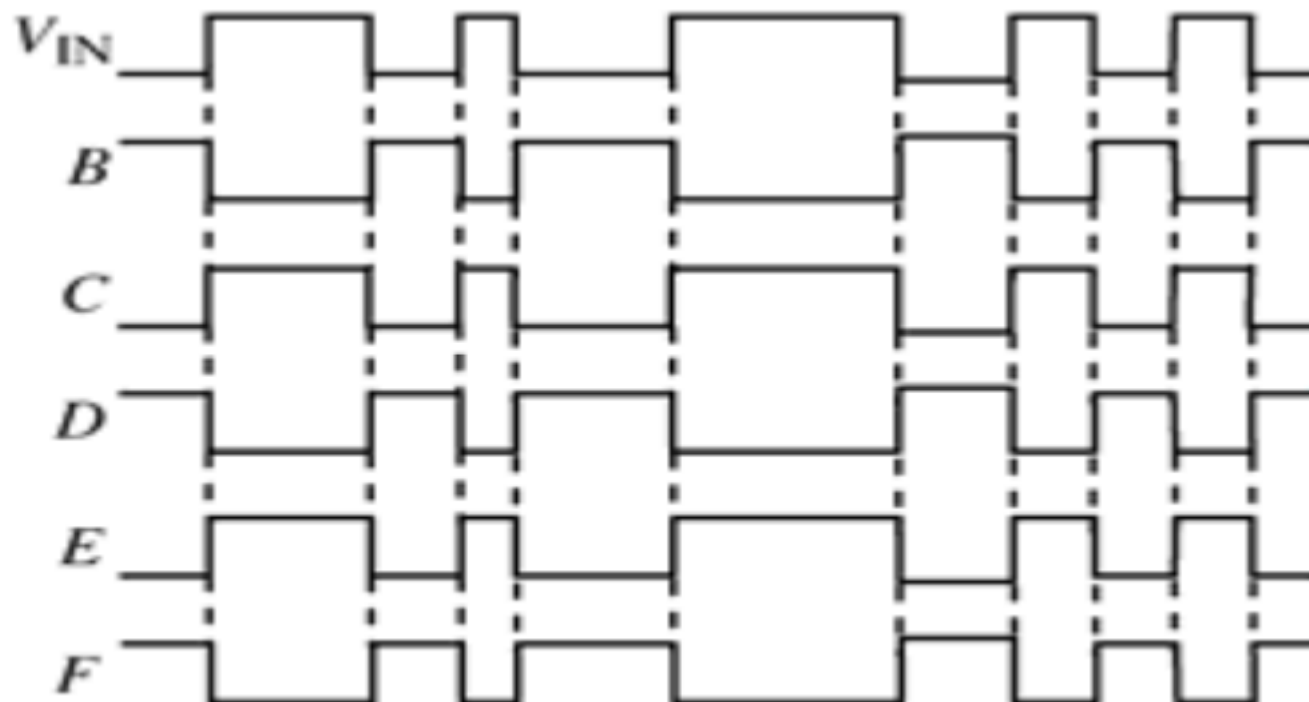
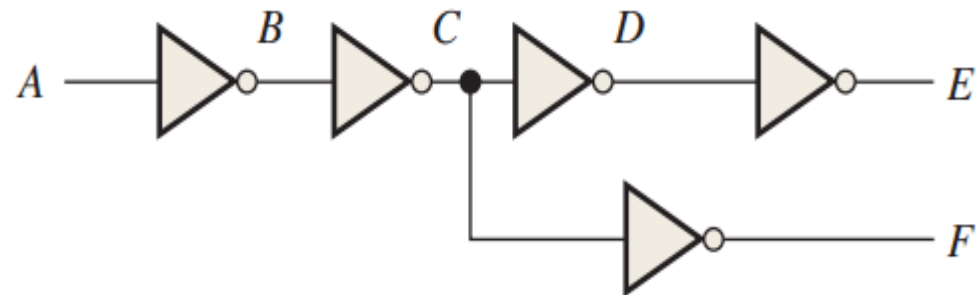
1. The input waveform shown in Figure 3-76 is applied to a system of two inverters connected in a series. Draw the output waveform across each inverter in proper relation to the input.



2. A combination of inverters is shown in Figure 3-77. If a LOW is applied to point A , determine the net output at points E and F .

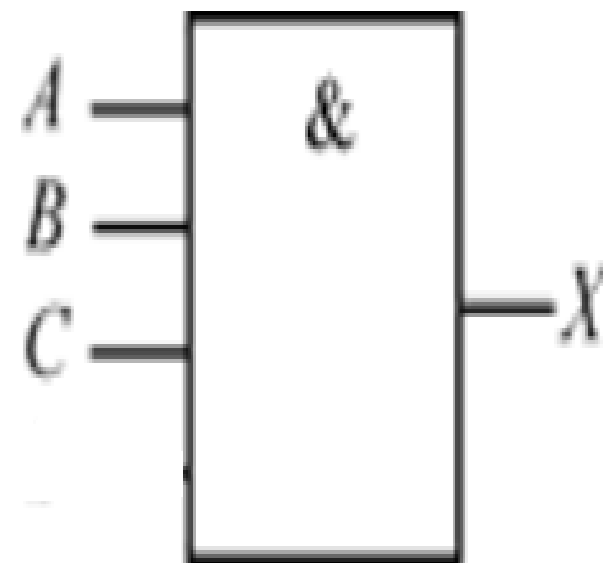


3- If the waveform in Figure 3–76 is applied to point **A** in Figure 3–77, determine the waveforms at points **B** through **F**.

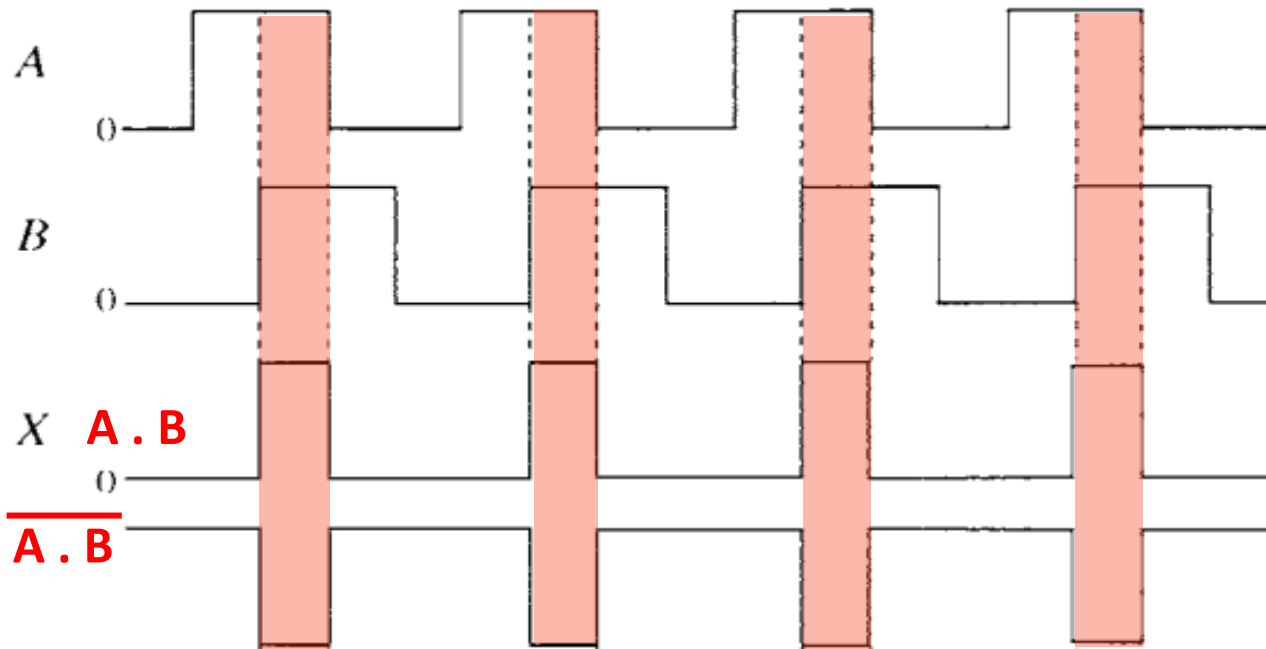


Section 3-2 The AND Gate

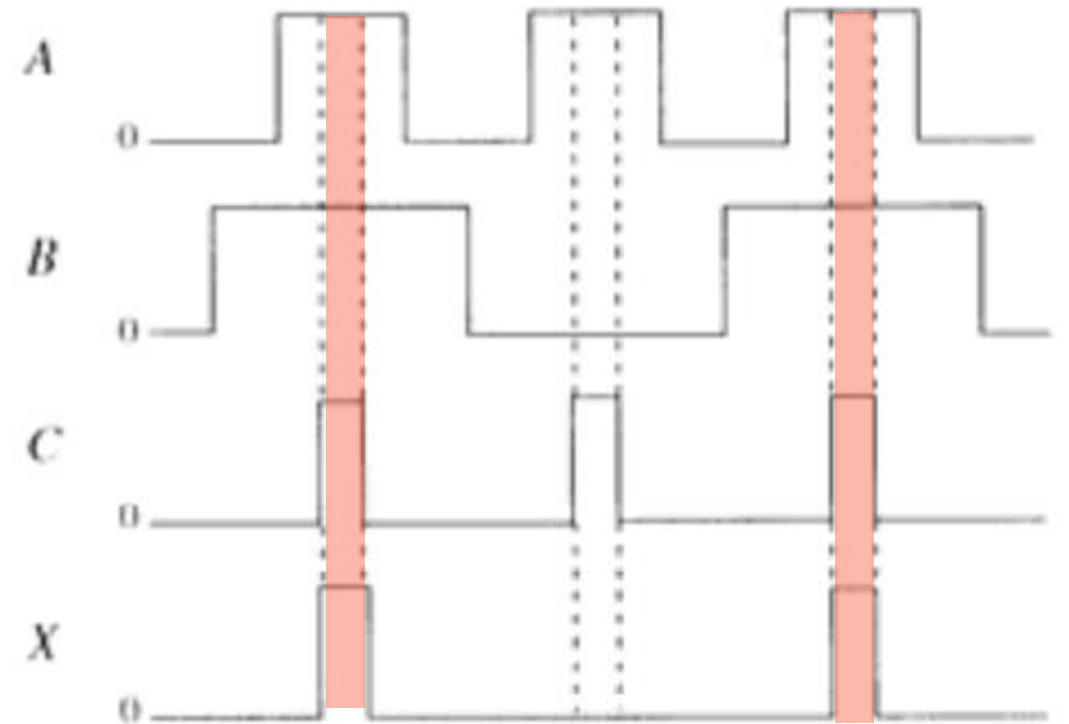
4- Draw the rectangular outline symbol for a 3-input AND gate.



6- The waveforms in Figure 3–79 are applied to points A and B of a 2-input AND gate followed by an inverter. Draw the output waveform.

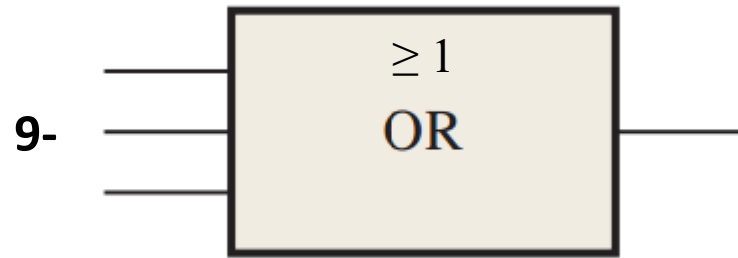


7- The input waveforms applied to a 3-input AND gate are as indicated in Figure 3–80. Show the output waveform in proper relation to the inputs with a timing diagram

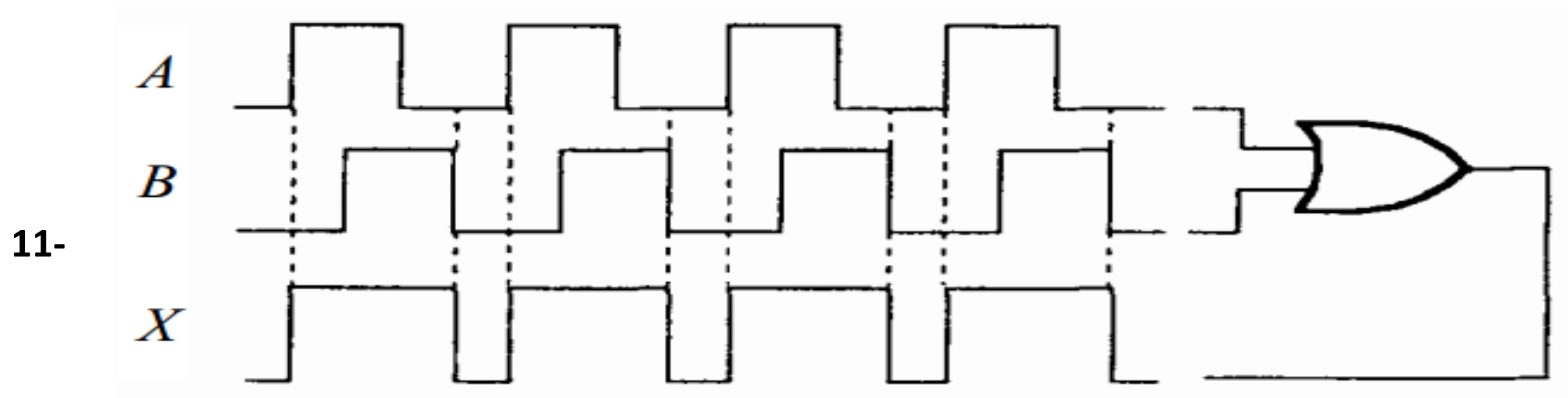


Section 3-3 The OR Gate

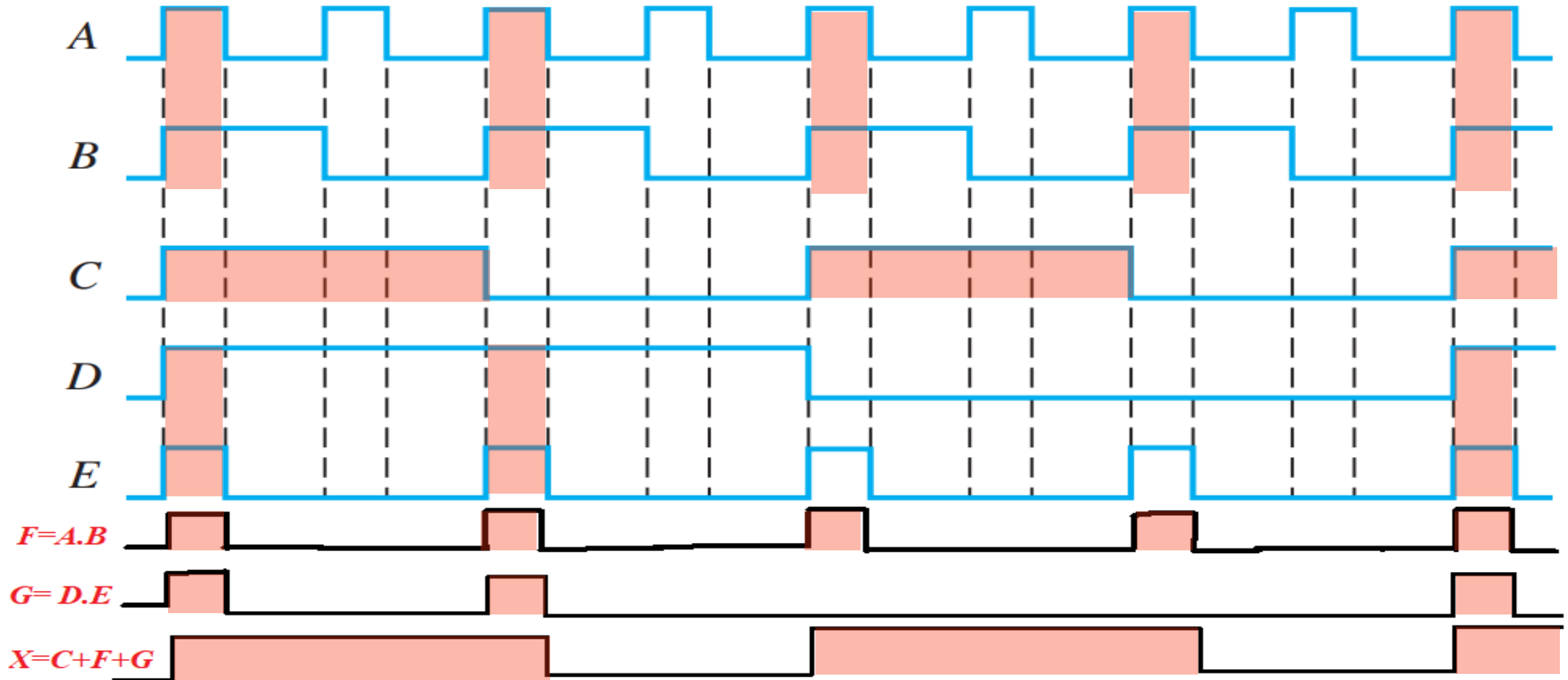
9. Draw the rectangular outline symbol for a 3-input OR gate.
10. Write the expression for a 4-input OR gate with inputs A , B , C , D , and output X .
11. Determine the output for a 2-input OR gate when the input waveforms are as in Figure 3-79 and draw a timing diagram.



10- $X = A + B + C + D$

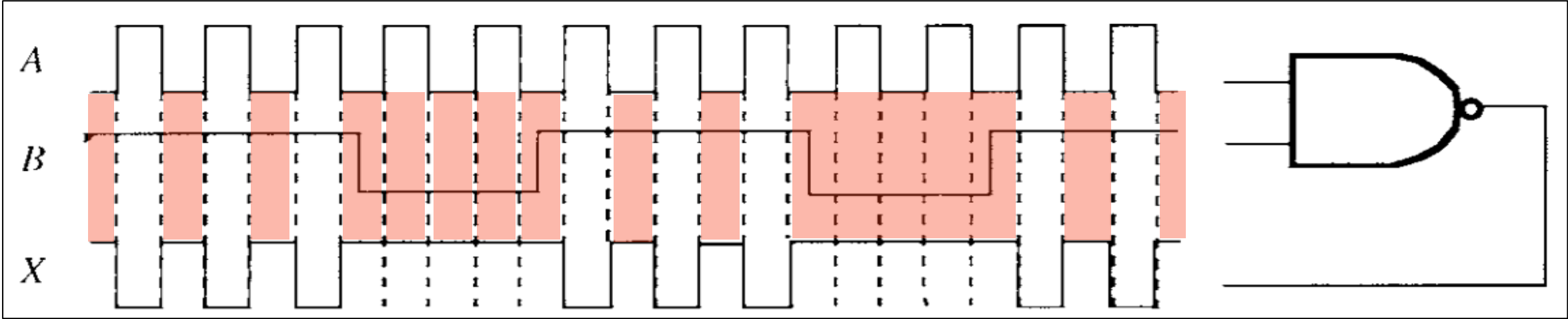


14- For the waveforms given in Figure 3–82, **A and B are ANDed with output F, D and E are ANDed with output G, and C, F, and G are ORed.** Draw the net output waveform.

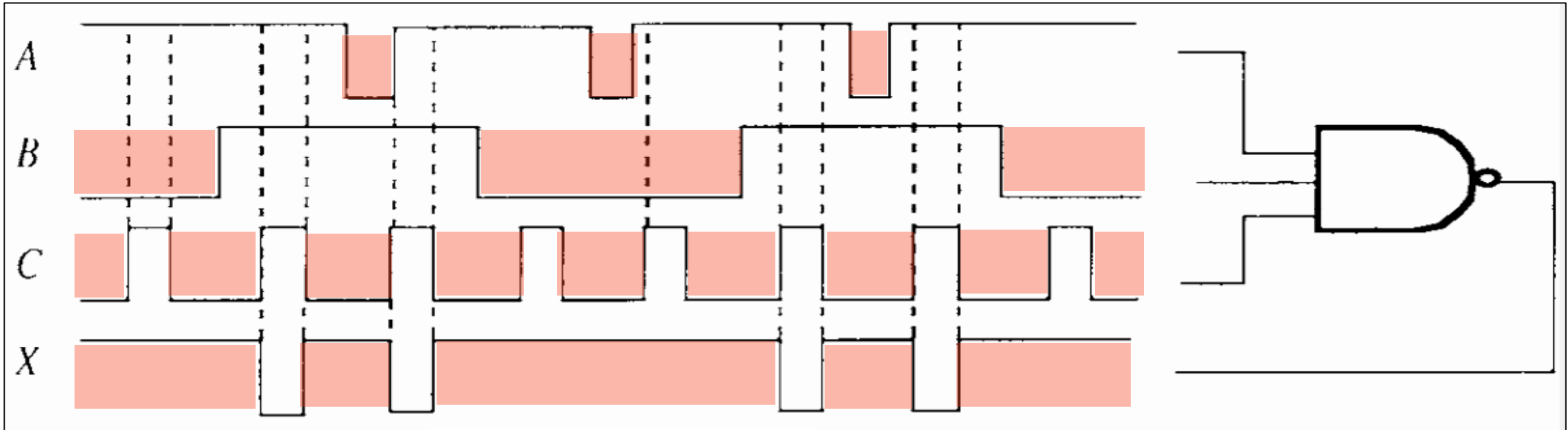


Section 3-4 The NAND Gate

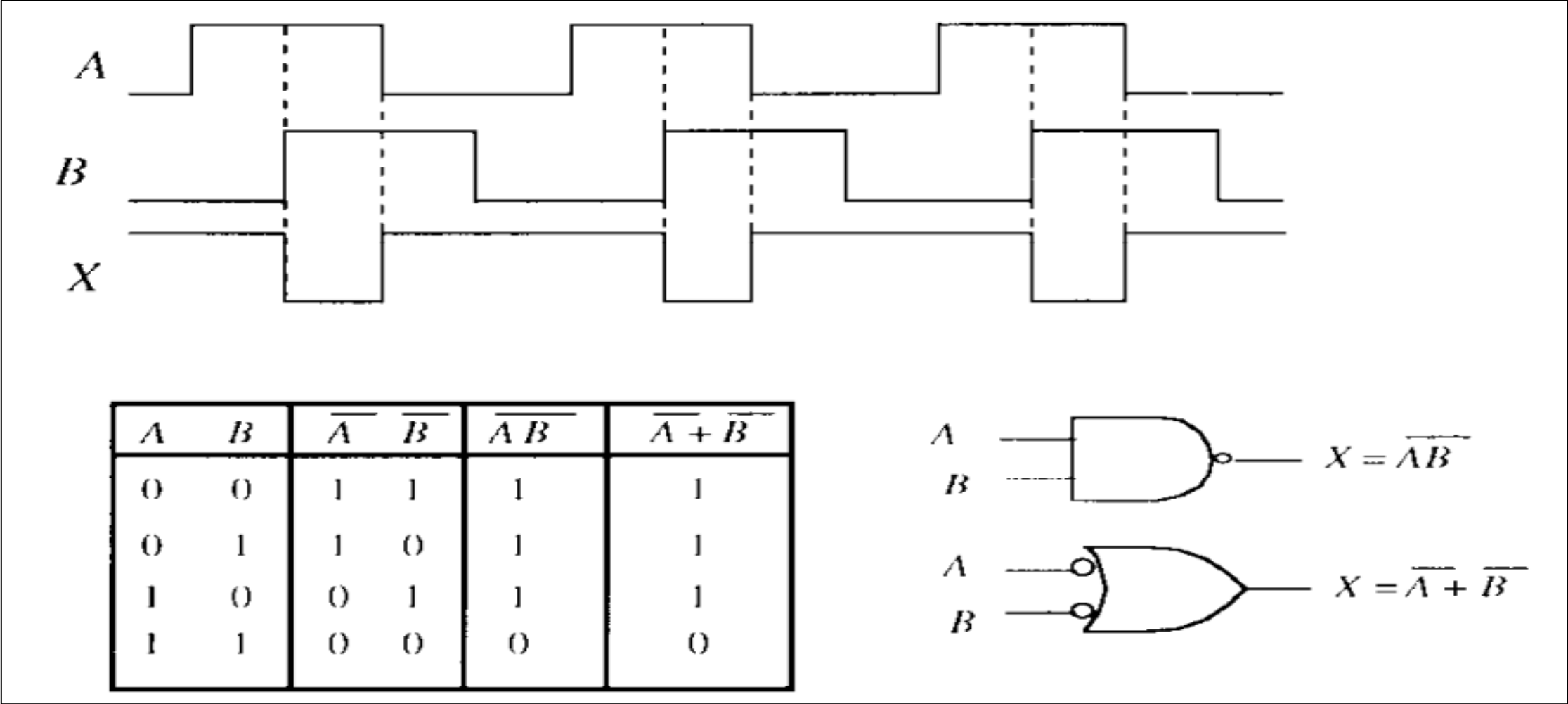
17. For the set of input waveforms in Figure 3-83, determine the output for the gate shown and draw the timing diagram.



18-

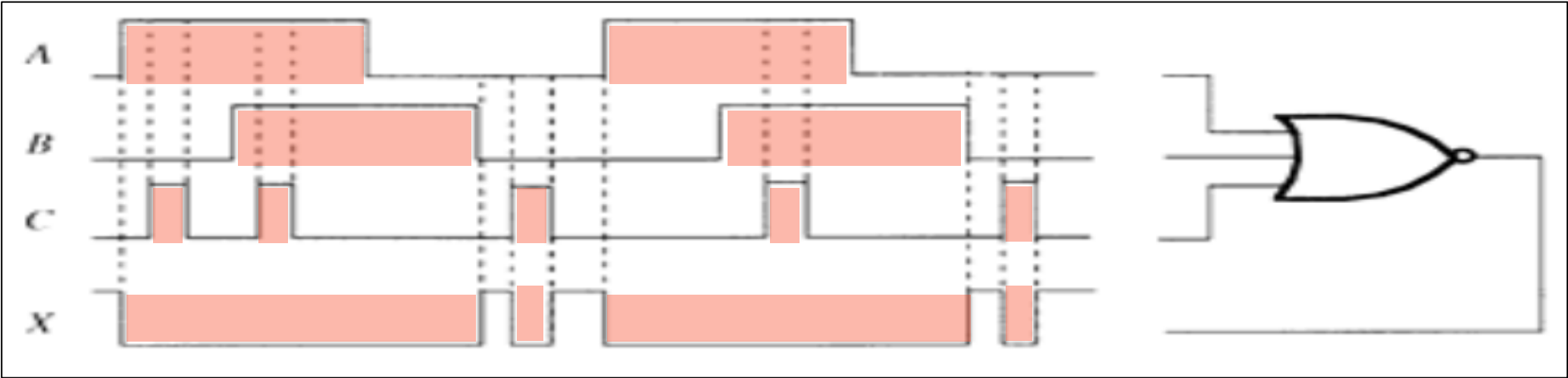


20- As you have learned, the two logic symbols shown in Figure 3–86 represent equivalent operations. The difference between the two is strictly from a functional viewpoint. For the NAND symbol, look for two HIGHS on the inputs to give a LOW output. For the negative-OR, look for at least one LOW on the inputs to give a HIGH on the output. Using these two functional viewpoints, show that each gate will produce the same output for the given inputs.

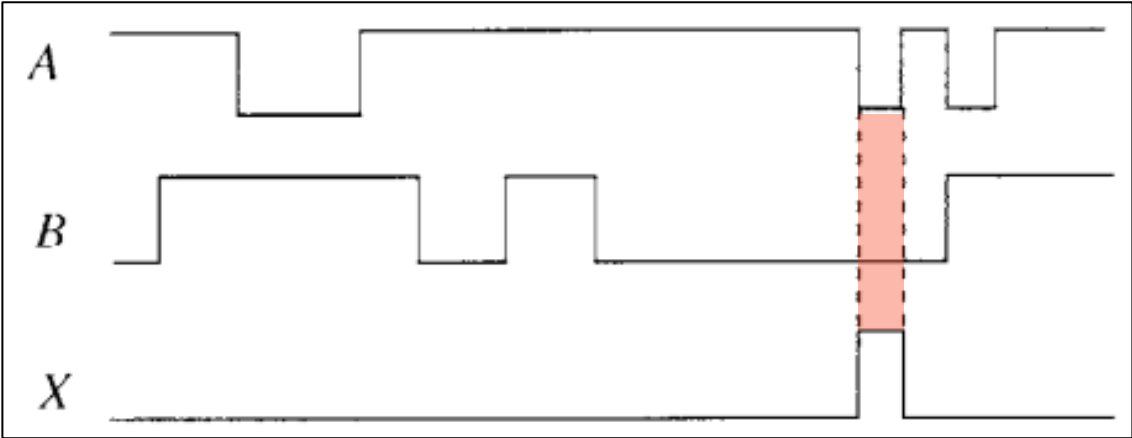


Section 3-5 The NOR Gate

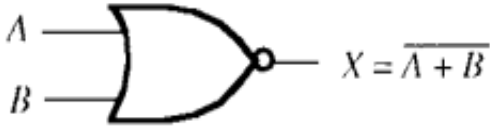
22 - Determine the output waveform in Figure 3-87 and draw the timing diagram.



24 - For the NOR symbol, look for at least one HIGH on the inputs to give a LOW on the output. For the negative-AND, look for two LOWs on the inputs to give a HIGH output. Using these two functional points of view, show that both gates in the Figure will produce the same output for the given inputs.



A	B	$\overline{A}$	$\overline{B}$	$\overline{A+B}$	$\overline{\overline{A}\overline{B}}$
0	0	1	1	1	1
0	1	1	0	0	0
1	0	0	1	0	0
1	1	0	0	0	0



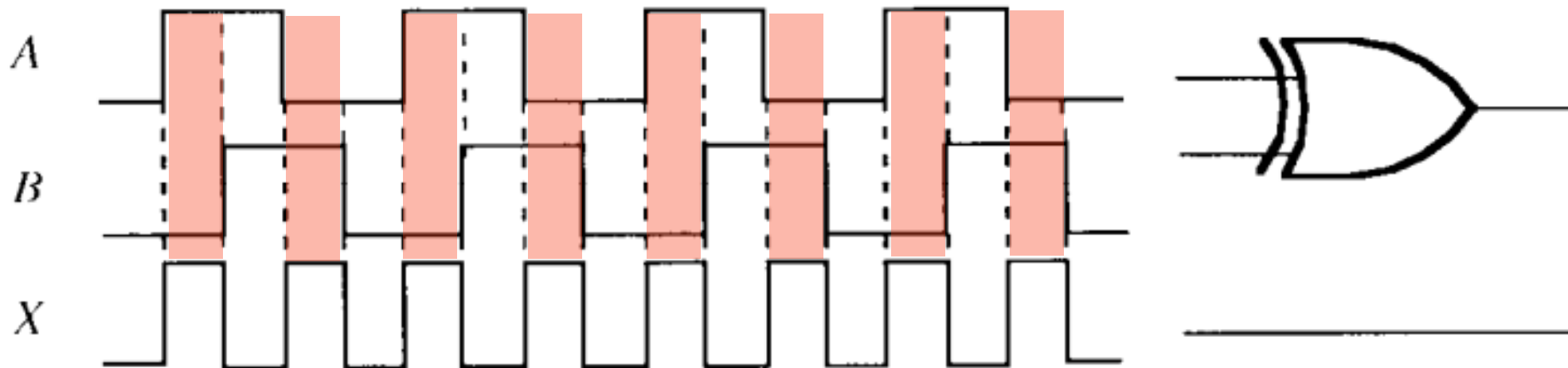
Section 3-6 The Exclusive-OR and Exclusive-NOR Gates

25- How does an exclusive-OR gate differ from an OR gate in its logical operation?

The output of the XOR gate is HIGH only when one input is HIGH. The output of the OR gate is HIGH any time one or more inputs are HIGH.

$$\begin{aligned} \text{XOR} &= \overline{A}B + A\overline{B} \\ \text{OR} &= A + B \end{aligned}$$

28- Determine the output of an exclusive-NOR gate for the inputs shown in Figure 3-79 and draw a timing diagram.



Section 3–8 Fixed-Function Logic Gates

33- In the comparison of certain logic devices, it is noted that the power dissipation for one type increases as the frequency increases. Is the device bipolar or CMOS?

- The power dissipation, P_D , of a logic gate is the product of the dc supply voltage and the average supply current.

$$P_D = V_{CC} \left(\frac{I_{CCH} + I_{CCL}}{2} \right)$$

, the supply current for the LOW output state as I_{CCL} and for the HIGH state as I_{CCH}

- CMOS gates have very low power dissipations compared to the bipolar family. the power dissipation of CMOS is dependent on the frequency (The power dissipation of CMOS increases with frequency). Power dissipation for bipolar gates is independent of frequency.

34 - Using the data sheets in Figures 3–65 and 3–66, determine the following:

- (a) 74LS00 power dissipation at maximum supply voltage and a 50% duty cycle
- (b) Minimum HIGH level output voltage for a 74LS00
- (c) Maximum propagation delay for a 74LS00
- (d) Maximum LOW level output voltage for a 74HC00A
- (e) Maximum propagation delay for a 74HC00A

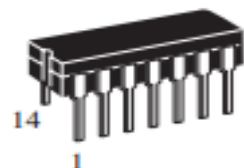
$$(a) \quad P = \left(\frac{I_{CCH} + I_{CCL}}{2} \right) V_{CC} = \left(\frac{1.6 \text{ mA} + 4.4 \text{ mA}}{2} \right) 5.5 \text{ V} = 16.5 \text{ mW}$$

QUAD 2-INPUT NAND GATE

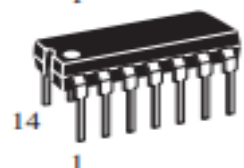
• ESD > 3500 Volts

SN54/74LS00

QUAD 2-INPUT NAND GATE
LOW POWER SCHOTTKY



J SUFFIX
CERAMIC
CASE 632-08



N SUFFIX
PLASTIC
CASE 646-06



D SUFFIX
SOIC
CASE 751A-02

ORDERING INFORMATION

SN54LSXXJ	Ceramic
SN74LSXXN	Plastic
SN74LSXXD	SOIC

SN54/74LS00

DC CHARACTERISTICS OVER OPERATING TEMPERATURE RANGE (unless otherwise specified)

Symbol	Parameter	Limits			Unit	Test Conditions
		Min	Typ	Max		
V_{IH}	Input HIGH Voltage	2.0			V	Guaranteed Input HIGH Voltage for All Inputs
V_{IL}	Input LOW Voltage	54		0.7	V	Guaranteed Input LOW Voltage for All Inputs
		74		0.8		
V_{IK}	Input Clamp Diode Voltage		-0.65	-1.5	V	$V_{CC} = \text{MIN}$, $I_{IN} = -18 \text{ mA}$
V_{OH}	Output HIGH Voltage	54	2.5	3.5	V	$V_{CC} = \text{MIN}$, $I_{OH} = \text{MAX}$, $V_{IN} = V_{IH}$ or V_{IL} per Truth Table
		74	2.7	3.5	V	
V_{OL}	Output LOW Voltage	54, 74	0.25	0.4	V	$I_{OL} = 4.0 \text{ mA}$ $V_{CC} = V_{CC} \text{ MIN}$, $V_{IN} = V_{IL}$ or V_{IH} per Truth Table $I_{OL} = 8.0 \text{ mA}$
		74	0.35	0.5	V	
I_{IH}	Input HIGH Current			20	μA	$V_{CC} = \text{MAX}$, $V_{IN} = 2.7 \text{ V}$
I_{IL}	Input LOW Current			0.1	mA	$V_{CC} = \text{MAX}$, $V_{IN} = 7.0 \text{ V}$
I_{IS}	Short Circuit Current (Note 1)			-0.4	mA	$V_{CC} = \text{MAX}$, $I_N = 0.4 \text{ V}$
I_{OS}	Short Circuit Current (Note 1)	-20		-100	mA	$V_{CC} = \text{MAX}$
I_{CC}	Power Supply Current Total, Output HIGH			1.6	mA	$V_{CC} = \text{MAX}$
	Total, Output LOW			4.4		

NOTE 1: Not more than one output should be shorted at a time, nor for more than 1 second.

AC CHARACTERISTICS ($T_A = 25^\circ\text{C}$)

Symbol	Parameter	Limits			Unit	Test Conditions
		Min	Typ	Max		
t_{PLH}	Turn-Off Delay, Input to Output		9.0	15	ns	$V_{CC} = 5.0 \text{ V}$ $C_L = 15 \text{ pF}$
t_{PHL}	Turn-On Delay, Input to Output		10	15	ns	

(b) Minimum HIGH level output voltage for a 74LS00 ($V_{OH (min)}$)= 2.7 v

(c) Maximum propagation delay for a 74LS00 (t_{PLH})= t_{PHL})= 15 ns

QUAD 2-INPUT NAND GATE

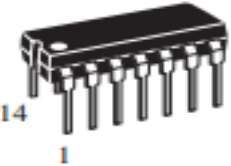
• ESD > 3500 Volts

SN54/74LS00

QUAD 2-INPUT NAND GATE
LOW POWER SCHOTTKY



J SUFFIX
CERAMIC
CASE 632-08



N SUFFIX
PLASTIC
CASE 646-06



D SUFFIX
SOIC
CASE 751A-02

ORDERING INFORMATION

SN54LSXXJ Ceramic

SN74LSXXN Plastic

SN74LSXXD SOIC

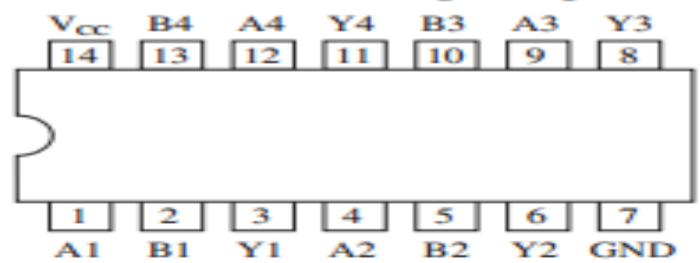
SN54/74LS00						
DC CHARACTERISTICS OVER OPERATING TEMPERATURE RANGE (unless otherwise specified)						
Symbol	Parameter	Limits			Unit	Test Conditions
		Min	Typ	Max		
V_{IH}	Input HIGH Voltage	2.0			V	Guaranteed Input HIGH Voltage for All Inputs
V_{IL}	Input LOW Voltage	54		0.7	V	Guaranteed Input LOW Voltage for All Inputs
		74		0.8		
V_{IK}	Input Clamp Diode Voltage		-0.65	-1.5	V	$V_{CC} = \text{MIN}$, $I_{IN} = -18 \text{ mA}$
V_{OH}	Output HIGH Voltage	54	2.5	3.5	V	$V_{CC} = \text{MIN}$, $I_{OH} = \text{MAX}$, $V_{IN} = V_{IH}$ or V_{IL} per Truth Table
		74	2.7	3.5	V	
V_{OL}	Output LOW Voltage	54, 74	0.25	0.4	V	$I_{OL} = 4.0 \text{ mA}$ $V_{CC} = V_{CC} \text{ MIN}$, $V_{IN} = V_{IL}$
		74	0.35	0.5	V	$I_{OL} = 8.0 \text{ mA}$ or V_{IH} per Truth Table
I_{IH}	Input HIGH Current			20	μA	$V_{CC} = \text{MAX}$, $V_{IN} = 2.7 \text{ V}$
I_{IL}	Input LOW Current			0.1	mA	$V_{CC} = \text{MAX}$, $V_{IN} = 7.0 \text{ V}$
I_{OS}	Short Circuit Current (Note 1)	-20		-100	mA	$V_{CC} = \text{MAX}$
I_{CC}	Power Supply Current Total, Output HIGH Total, Output LOW			1.6	mA	$V_{CC} = \text{MAX}$
				4.4		

NOTE 1: Not more than one output should be shorted at a time, nor for more than 1 second.

AC CHARACTERISTICS ($T_A = 25^\circ\text{C}$)					
Symbol	Parameter	Limits			Unit
		Min	Typ	Max	
t_{PLH}	Turn-Off Delay, Input to Output		9.0	15	ns
t_{PHL}	Turn-On Delay, Input to Output		10	15	ns

(d) Maximum LOW level output voltage for a 74HC00A ($V_{OL(max)}$)= .4 v

Pinout: 14–Load Packages (Top View)



FUNCTION TABLE		
Inputs		Output
A	B	Y
L	L	H
L	H	H
H	L	H
H	H	L

DC CHARACTERISTICS (Voltages Referenced to GND)

MC54/74HC00A

Symbol	Parameter	Condition	V _{CC} V	Guaranteed Limit			Unit
				–55 to 25°C	≤85°C	≤125°C	
V _{IH}	Minimum High-Level Input Voltage	V _{out} = 0.1 V or V _{CC} – 0.1 V I _{out} ≤ 20 μA	2.0	1.50	1.50	1.50	V
			3.0	2.10	2.10	2.10	
			4.5	3.15	3.15	3.15	
			6.0	4.20	4.20	4.20	
V _{IL}	Maximum Low-Level Input Voltage	V _{out} = 0.1 V or V _{CC} – 0.1 V I _{out} ≤ 20 μA	2.0	0.50	0.50	0.50	V
			3.0	0.90	0.90	0.90	
			4.5	1.35	1.35	1.35	
			6.0	1.80	1.80	1.80	
V _{OH}	Minimum High-Level Output Voltage	V _{in} = V _{IH} or V _{IL} I _{out} ≤ 20 μA	2.0	1.9	1.9	1.9	V
			4.5	4.4	4.4	4.4	
		V _{in} = V _{IH} or V _{IL} I _{out} ≤ 2.4 mA I _{out} ≤ 4.0 mA I _{out} ≤ 5.2 mA	3.0	2.48	2.34	2.20	
			4.5	3.98	3.84	3.70	
V _{OL}	Maximum Low-Level Output Voltage	V _{in} = V _{IH} or V _{IL} I _{out} ≤ 20 μA	2.0	0.1	0.1	0.1	V
			4.5	0.1	0.1	0.1	
			6.0	0.1	0.1	0.1	
		V _{in} = V _{IH} or V _{IL} I _{out} ≤ 2.4 mA I _{out} ≤ 4.0 mA I _{out} ≤ 5.2 mA	3.0	0.26	0.33	0.40	
			4.5	0.26	0.33	0.40	
			6.0	0.26	0.33	0.40	
I _{in}	Maximum Input Leakage Current	V _{in} = V _{CC} or GND	6.0	±0.1	±1.0	±1.0	μA
I _{CC}	Maximum Quiescent Supply Current (per Package)	V _{in} = V _{CC} or GND I _{out} = 0 μA	6.0	1.0	10	40	μA

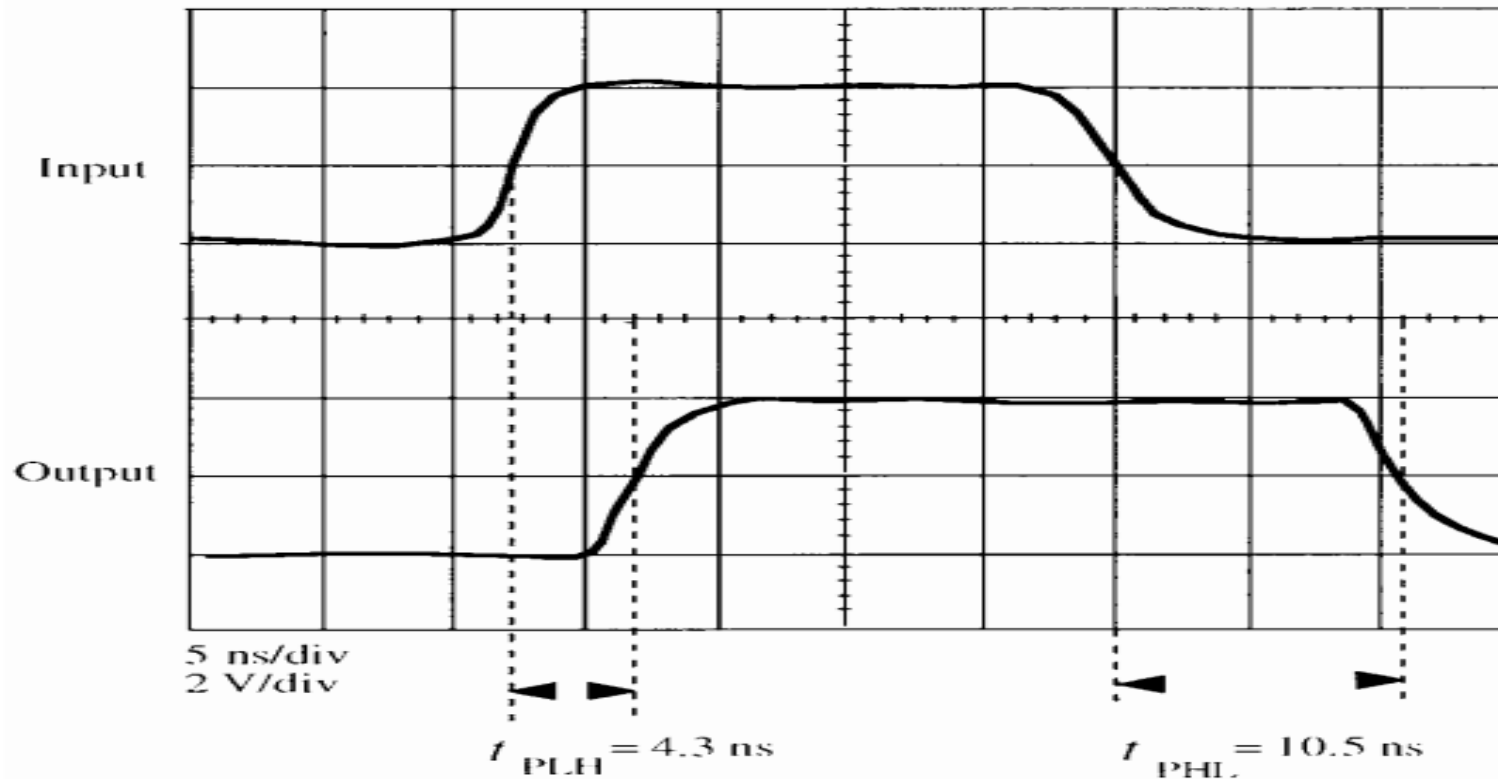
(e) Maximum propagation delay for a 74HC00A :

@ $V_{CC} = 2 \text{ V}$, $t_{PHL} = t_{PLH} = 75 \text{ ns}$; @ $V_{CC} = 6 \text{ V}$, $t_{PHL} = t_{PLH} = 13 \text{ ns}$

AC CHARACTERISTICS ($C_L = 50 \text{ pF}$, Input $t_r = t_f = 6 \text{ ns}$)

Symbol	Parameter	V_{CC} V	Guaranteed Limit			Unit
			-55 to 25°C	≤85°C	≤125°C	
t_{PLH} , t_{PHL}	Maximum Propagation Delay, Input A or B to Output Y	2.0	75	95	110	ns
		3.0	30	40	55	
		4.5	15	19	22	
		6.0	13	16	19	
t_{TLH} , t_{THL}	Maximum Output Transition Time, Any Output	2.0	75	95	110	ns
		3.0	27	32	36	
		4.5	15	19	22	
		6.0	13	16	19	
C_{in}	Maximum Input Capacitance		10	10	10	pF

35- Determine t_{PLH} and t_{PHL} from the oscilloscope display in Figure 3–91. The readings indicate volts/div and sec/div for each channel.



36- Gate A has $t_{PLH} = t_{PHL} = 6$ ns. Gate B has $t_{PLH} = t_{PHL} = 10$ ns. Which gate can be operated at a higher frequency?

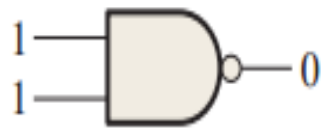
➤ Gate A can be operated at the highest frequency because it has shorter propagation delay times than Gate B.

37- If a logic gate operates on a dc supply voltage of +5 V and draws an average current of 4 mA, what is its power dissipation?

➤ $P_D = V_{CC} \cdot I_{CC} = (5 \text{ V})(4 \text{ mA}) = 20 \text{ mW}$

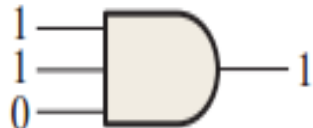
Section 3-9 Troubleshooting

39. Examine the conditions indicated in Figure 3-92, and identify the faulty gates.



(a)

true



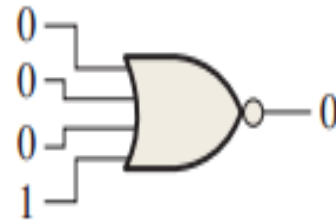
(b)

False



(c)

False



(d)

true



(e)

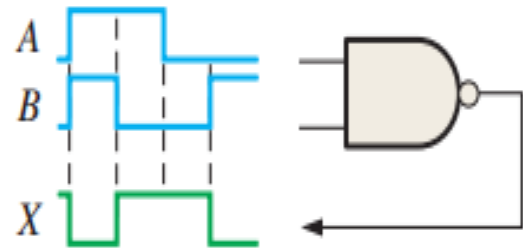
False



(f)

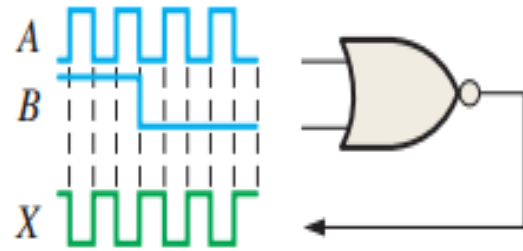
true

40. Determine the faulty gates in Figure 3–93 by analyzing the timing diagrams.



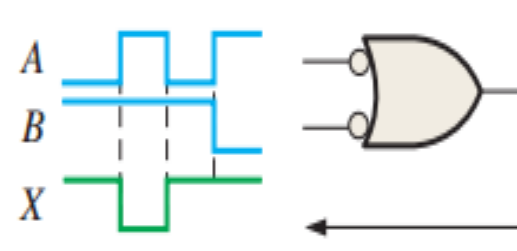
(a)

False



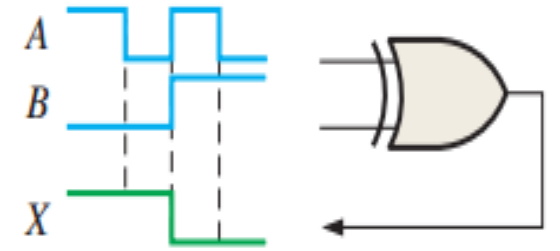
(b)

False



(c)

True



(d)

False

The end of sheet 3

Thank You